Prodigy Anesthesia Quick Review

Chronic Renal Failure

What are the two leading causes of end-stage renal disease?	Diabetes mellitus and hypertension. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 341.*
What percentage of the nephrons will be destroyed before patients with ESRD become symptomatic?	90% *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 341.*
What electrolyte imbalances does chronic renal failure cause?	Hyponatremia, hyperkalemia, hyperphosphatemia, hypocalcemia, and metabolic acidosis. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 651.*
What are the clinical signs of chronic renal insufficiency?	Initially, nonspecific signs such as malaise and fatigue may be present. Volume overload, electrolyte and acid-base alterations, and decreased urine output are late signs of chronic renal insufficiency. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 342-344.*
What are the first substances that begin to accumulate in the plasma as renal failure progresses?	Urea and creatinine are the first substances to accumulate in the plasma in patients with ESRD. They begin to elevate when renal function drops to 50%. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 342.*
When during the progression of renal failure does the serum potassium begin to rise? How does the body try to compensate for this?	When the glomerular filtration reaches 10% of normal, serum potassium begins to elevate. The kidneys compensate by increasing blood flow through the remaining nephrons and by excreting more potassium through the gastrointestinal tract. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 342.*
What hematologic changes occur with ESRD?	Anemia is universal among patients with ESRD and is due to chronic blood loss, hemolysis, uremic suppression of bone marrow, iron and folate deficiency, and a decrease in the production of erythropoetin. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 652.*

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How does chronic renal failure affect coagulation?	Patients with ESRD exhibit prolonged bleeding times due to uremia despite normal platelet counts, prothrombin time, and partial thromboplastin time. They may also exhibit a predisposition to thromboembolism related to impaired fibrinolysis, increased platelet aggregation, and altered protein C and protein S activity. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 6th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2008: 824.*
What are the anesthesia considerations at induction of anesthesia for patients with chronic renal failure?	Induction should proceed slowly using conventional agents (thiopental, propofol, etomidate) to avoid excessive hypotension. Regardless of the volume status of the patient with ESRD, they respond as if they were hypovolemic. Uremia can slow gastric emptying time, which can increase the risk for gastric reflux. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 348.*
What positioning concerns would you have for a patient with chronic renal failure?	Poor nutritional status makes their skin extremely susceptible to sloughing and bruising. Be sure to provide extra padding at vulnerable areas such as the dialysis fistulas, elbows, knees, and heels. Try to avoid tucking the arm with the fistula if the patient undergoes dialysis so that it can be evaluated during the procedure. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 338.*
What is of extreme importance when performing neuraxial anesthesia or placing invasive lines in patients with chronic renal failure?	Aseptic technique should be adhered to strictly because these patients are extremely susceptible to infection. *Stoelting RK, Dierdorf SF. Anesthesia & Co-Existing Diseases. 5th ed. New York, NY: Churchill-Livingston; 2008: 339.*
Would you expect awake patients with chronic renal failure to exhibit hyperventilation or hypoventilation at rest? Why or why not?	They tend to exhibit hyperventilation. This may be due to chronic metabolic acidosis and the increased predisposition to pulmonary edema. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1411.*
A patient with chronic renal failure exhibits extreme hyperkalemia. What are appropriate methods for reducing the potassium level urgently?	Severe hyperkalemia can be life-threatening and must be treated immediately. Administration of calcium chloride stabilizes the heart against dangerous arrhythmias while other methods of reducing the serum potassium are begun. Administration of glucose and insulin, sodium bicarbonate administration, hyperventilation, dialysis, and administration of potassium-binding resins are all methods of reducing the serum potassium level. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1129-1130.*

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Should patients on dialysis undergo treatment during the perioperative period?	Yes. Patients who are on hemodialysis should be dialyzed within 24 hours of surgery to minimize the chances of volume overload, hyperkalemia, and uremic bleeding. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 346.*
What is the most common cause of death in patients receiving hemodialysis?	Cardiovascular events and infection *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 345.*
What is the most common adverse event in patients receiving hemodialysis?	hypotension *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 345.*
What pathologic conditions are associated with uremia?	Cardiac enlargement, ventricular hypertrophy, CHF, impaired mental acuity, and immunocompromise. *Yao FF. Anesthesiology: Problem-Oriented Patient Management. 7th ed. Philadelphia: Lippincott, Williams, and Wilkins; 2012: 654-655.*
How is osteodystrophy linked to chronic renal failure?	As the kidneys fail, there is an accumulation of phosphate. The elevated phosphate levels lead to a decrease in serum calcium levels. Hypocalcemia causes the release of parathyroid hormone which causes bone demineralization to restore serum calcium levels. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 342-343.*
Is succinylcholine contraindicated in patients with chronic renal failure?	No. Patients with chronic renal failure do not exhibit exaggerated hyperkalemic responses to succinylcholine. If the preoperative potassium is in the high to high-normal range, you should evaluate whether the increase in potassium due to succinylcholine (0.5-1.0 mEq/L) might predispose the patient to dysrhythmias. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 347.*
Why do patients with chronic renal failure require smaller doses of thiopental?	Patients with chronic renal disease exhibit reduced plasma protein binding of drugs. As a result, the free fraction of thiopental is increased dramatically. The IV dose for thiopental is generally half that given for patients with normal renal function. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1411.*

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How does the pharmacokinetic profile of propofol in patients with ESRD compare to patients with normal renal function? What about ketamine? Etomidate?

Propofol is rapidly metabolized by the liver and does not show signs of prolonged duration of action in patients with ESRD. Ketamine and etomidate do not undergo as extensive a degree of protein binding as that seen with thiopental, but they do have a slightly increased free fraction.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1411.

Why is the duration of action of midazolam prolonged in patients with renal disease?

Benzodiazepines are extensively protein bound. The reduced protein binding in patients with renal disease results in an increased free fraction of the drug. Also, 60-80% of midazolam is excreted as an active metabolite. In renal disease, this metabolite accumulates and can result in prolonged sedation.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1411.

Which three muscle relaxants exhibit the least degree of prolongation in patients with renal failure?

Only succinylcholine, atracurium, and cisatracurium do not rely on renal excretion and have minimal if any prolongation of effect in patients with renal disease. Laudanosine, the principal metabolite of atracurium and cisatracurium, does rely on renal excretion. It produces no neuromuscular relaxant effect, but can produce CNS excitation if it accumulates. There are no clinical data linking laudanosine with complications with ESRD patients, but careful titration using a peripheral nerve stimulator would be prudent.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 347.

How does single-dose administration of morphine compare to continuous infusion or repeated dosing of morphine in patients with renal disease?

A single dose of morphine demonstrates no significant prolongation of effect in patients with renal disease. With repeated dosing or infusion, however, morphine's active metabolite, morphine-6 glucuronide, accumulates and can result in prolonged narcosis.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1412.

How are the pharmacokinetics of fentanyl and remifentanil affected in patients with renal disease?

Fentanyl has no active metabolites, no alteration in its free fraction, and is tolerated well in small, titrated doses in patients with renal impairment. Remifentanil undergoes extensive metabolism in the blood and tissue by esterases and is not affected to any significant degree by renal disease.

Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1412.

How is the duration of action of anticholinesterase drugs affected by chronic renal failure?

Approximately 50% of neostigmine undergoes renal elimination and 75% of edrophonium and pyridostigmine undergo renal elimination. As a result, the elimination half-times of these drugs is significantly prolonged by renal failure.

Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 348.

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